The omega-groupoid laws for path concatenation and inversion: unit laws, inverse laws, involution, and associativity

The Univalent Foundations Program

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Editorial note. The counted statement is the author lemma labelled thm:omg (source lines 393-404), an enumerate of four laws with labels item:omg1 to item:omg4 retained verbatim.

Editorial note. The proof span 410-506 is the author's single proof environment for the lemma, retained line for line; it contains per-item first and second proofs as prose items.

Editorial note. The statement and proof are self-contained: all cited labels (the item labels and the lemma's own label) are defined inside the retained material.

Editorial scope. One original result of the HoTT book chapter cha:basics, counted once: the omega-groupoid structure laws for path concatenation and inversion. The counted lemma and its single author proof environment (which itself presents a first and a concise second argument per item) are retained; the surrounding topological commentary remains uncounted in the source.

Allowed standard prerequisites (editorial).

- Path concatenation p concat q and path inversion (opposite p) as operations on the identity types of a type A, each with its definitional behaviour on refl.
- The induction principle for identity types (path induction) with its conversion rule: to prove a statement about all paths it suffices to prove it for refl, and the proof term then computes on refl by definition.
- Iterated identity types: the laws asserted are themselves propositional equalities, that is, elements of identity types of identity types (paths of paths).

Lemma 0.0.1. *Suppose* $A : \mathcal{U}$*, that* $x, y, z, w : A$ *and that* $p : x = y$ *and* $q : y = z$ *and* $r : z = w$*. We have the following:*

(i) $p = p \cdot \mathsf{refl}_y$ *and* $p = \mathsf{refl}_x \cdot p$.
(ii) $p^{-1} \cdot p = \mathsf{refl}_y$ *and* $p \cdot p^{-1} = \mathsf{refl}_x$.
(iii) $(p^{-1})^{-1} = p$.

(iv) $p \cdot (q \cdot r) = (p \cdot q) \cdot r$.

Proof of Lemma 0.0.1. All the proofs use the induction principle for equalities.

(i) *First proof:* let $D : \prod_{(x,y:A)}(x = y) \to \mathcal{U}$ be the type family given by

$$D(x, y, p) :\equiv (p = p \cdot \mathsf{refl}_y).$$

Then $D(x, x, \mathsf{refl}_x)$ is $\mathsf{refl}_x = \mathsf{refl}_x \cdot \mathsf{refl}_x$. Since $\mathsf{refl}_x \cdot \mathsf{refl}_x \equiv \mathsf{refl}_x$, it follows that $D(x, x, \mathsf{refl}_x) \equiv (\mathsf{refl}_x = \mathsf{refl}_x)$. Thus, there is a function

$$d :\equiv \lambda x.\, \mathsf{refl}_{\mathsf{refl}_x} : \prod_{x:A} D(x, x, \mathsf{refl}_x).$$

Now the induction principle for identity types gives an element $\mathsf{ind}_{=_A}(D, d, x, y, p) : (p = p \cdot \mathsf{refl}_y)$ for each $p : x = y$. The other equality is proven similarly.

Second proof: by induction on p, it suffices to assume that y is x and that p is refl_x. But in this case, we have $\mathsf{refl}_x \cdot \mathsf{refl}_x \equiv \mathsf{refl}_x$.

(ii) *First proof:* let $D : \prod_{(x,y:A)}(x = y) \to \mathcal{U}$ be the type family given by

$$D(x, y, p) :\equiv (p^{-1} \cdot p = \mathsf{refl}_y).$$

Then $D(x, x, \mathsf{refl}_x)$ is ${\mathsf{refl}_x}^{-1} \cdot \mathsf{refl}_x = \mathsf{refl}_x$. Since ${\mathsf{refl}_x}^{-1} \equiv \mathsf{refl}_x$ and $\mathsf{refl}_x \cdot \mathsf{refl}_x \equiv \mathsf{refl}_x$, we get that $D(x, x, \mathsf{refl}_x) \equiv (\mathsf{refl}_x = \mathsf{refl}_x)$. Hence we find the function

$$d :\equiv \lambda x.\, \mathsf{refl}_{\mathsf{refl}_x} : \prod_{x:A} D(x, x, \mathsf{refl}_x).$$

Now path induction gives an element $\mathsf{ind}_{=_A}(D, d, x, y, p) : p^{-1} \cdot p = \mathsf{refl}_y$ for each $p : x = y$ in A. The other equality is similar.

Second proof: by induction, it suffices to assume p is refl_x. But in this case, we have $p^{-1} \cdot p \equiv {\mathsf{refl}_x}^{-1} \cdot \mathsf{refl}_x \equiv \mathsf{refl}_x$.

(iii) *First proof:* let $D : \prod_{(x,y:A)}(x = y) \to \mathcal{U}$ be the type family given by

$$D(x, y, p) :\equiv ((p^{-1})^{-1} = p).$$

Then $D(x, x, \mathsf{refl}_x)$ is the type $(({\mathsf{refl}_x}^{-1})^{-1} = \mathsf{refl}_x)$. But since ${\mathsf{refl}_x}^{-1} \equiv \mathsf{refl}_x$ for each $x : A$, we have $({\mathsf{refl}_x}^{-1})^{-1} \equiv {\mathsf{refl}_x}^{-1} \equiv \mathsf{refl}_x$, and thus $D(x, x, \mathsf{refl}_x) \equiv (\mathsf{refl}_x = \mathsf{refl}_x)$. Hence we find the function

$$d :\equiv \lambda x.\, \mathsf{refl}_{\mathsf{refl}_x} : \prod_{x:A} D(x, x, \mathsf{refl}_x).$$

Now path induction gives an element $\mathsf{ind}_{=_A}(D, d, x, y, p) : (p^{-1})^{-1} = p$ for each $p : x = y$.

Second proof: by induction, it suffices to assume p is refl_x. But in this case, we have $(p^{-1})^{-1} \equiv ({\mathsf{refl}_x}^{-1})^{-1} \equiv \mathsf{refl}_x$.

(iv) *First proof:* let $D_1 : \prod_{(x,y:A)}(x = y) \to \mathcal{U}$ be the type family given by

$$D_1(x, y, p) :\equiv \prod_{(z,w:A)} \prod_{(q:y=z)} \prod_{(r:z=w)} \big(p \cdot (q \cdot r) = (p \cdot q) \cdot r\big).$$

Then $D_1(x, x, \mathsf{refl}_x)$ is

$$\prod_{(z,w:A)} \prod_{(q:x=z)} \prod_{(r:z=w)} \big(\mathsf{refl}_x \cdot (q \cdot r) = (\mathsf{refl}_x \cdot q) \cdot r\big).$$

To construct an element of this type, let $D_2 : \prod_{(x,z:A)}(x = z) \to \mathcal{U}$ be the type family

$$D_2(x, z, q) :\equiv \prod_{(w:A)} \prod_{(r:z=w)} \big(\mathsf{refl}_x \cdot (q \cdot r) = (\mathsf{refl}_x \cdot q) \cdot r\big).$$

Then $D_2(x, x, \mathsf{refl}_x)$ is

$$\prod_{(w:A)} \prod_{(r:x=w)} \big(\mathsf{refl}_x \cdot (\mathsf{refl}_x \cdot r) = (\mathsf{refl}_x \cdot \mathsf{refl}_x) \cdot r\big).$$

To construct an element of *this* type, let $D_3 : \prod_{(x,w:A)}(x = w) \to \mathcal{U}$ be the type family

$$D_3(x, w, r) :\equiv \big(\mathsf{refl}_x \cdot (\mathsf{refl}_x \cdot r) = (\mathsf{refl}_x \cdot \mathsf{refl}_x) \cdot r\big).$$

Then $D_3(x, x, \mathsf{refl}_x)$ is

$$\big(\mathsf{refl}_x \cdot (\mathsf{refl}_x \cdot \mathsf{refl}_x) = (\mathsf{refl}_x \cdot \mathsf{refl}_x) \cdot \mathsf{refl}_x\big)$$

which is definitionally equal to the type $(\mathsf{refl}_x = \mathsf{refl}_x)$, and is therefore inhabited by $\mathsf{refl}_{\mathsf{refl}_x}$. Applying the path induction rule three times, therefore, we obtain an element of the overall desired type.

Second proof: by induction, it suffices to assume p, q, and r are all refl_x. But in this case, we have

$$\begin{aligned} p \cdot (q \cdot r) &\equiv \mathsf{refl}_x \cdot (\mathsf{refl}_x \cdot \mathsf{refl}_x) \\ &\equiv \mathsf{refl}_x \\ &\equiv (\mathsf{refl}_x \cdot \mathsf{refl}_x) \cdot \mathsf{refl}_x \\ &\equiv (p \cdot q) \cdot r. \end{aligned}$$

Thus, we have $\mathsf{refl}_{\mathsf{refl}_x}$ inhabiting this type. □